

THE INTERNET OF THINGS



News, views and development on the Internet of Things

“IoT helps provide new services that might result into streams of revenue”

Tata Elxsi offers solutions for enabling the Internet of Things (IoT) in the areas of smart energy, transportation and remote healthcare, as well as industry-leading wireless access and backhaul solutions (on GSM, 3G, WiMAX, LTE and WiFi), several low-power devices for challenging outdoor deployments, and networking appliances for network automation and resource management.

Shyam Ananthnarayan, vice president-marketing, Embedded Product Design, and Biswajit Biswas, head-strategy, Wireless Communication, Tata Elxsi, explain to Janani Gopalakrishnan Vikram the changing definition of the IoT, and the challenges in using existing communication infrastructure for the same

What is the current scope of the IoT?

The communication paradigm continues to shift from people-to-people to people-to-machine, and now machine-to-machine. The challenge IoT brings forth for the network is that devices that will generate the Internet traffic are no longer going to be intelligent computers, but ‘things.’ These can include cars, home automation sensors, surveillance cameras mounted on the poles on highways, containers sitting in cargo trucks on transit, your future unique identification card, health monitoring devices or even your passport. The IoT will help to realise the dream of living in a smart city where people will live in smart homes.

With the increasing pressure on traditional service providers to monetise the network infrastructure, the IoT helps in providing new services that might result into streams of revenue. One of the significant trends is the decline of the ARPU-based voice traffic as most of the traffic increase is on the data side. This is further poised to grow with innovation and rapid growth of the IoT technology.

Also, it depends not alone on technology but equally on how the

government, society and individuals participate in embracing this massively impacting phenomenon called IoT.

What are the challenges involved in using existing communication networks for IoT applications? Could you tell us, at a broad level, how these can be overcome?

Existing long-range communication technology, especially wireless access standards, have not been designed keeping machine communication in mind. These are designed for voice and high-volume data communication. Machine communication data traffic is short and bursty in nature, and devices usually are battery-powered, which necessitates stringent power management for link control and signalling. For the time being, 3G and 4G (WiMax, LTE) may be good enough, but there are rising needs for a different access technology suiting the emerging IoT needs.

Other than a light-weight open wireless standard as the backbone, there is going to be a need for more and more spectrum. Currently, all the spectrums available are in multiples of gigahertz range. This won't be suitable for IoT communication, the

reason being poor NLOS performance and higher power requirement. Hence it would be necessary to get more affordable spectrums in sub-gigahertz bands. Spectrum known as TVWS (TV Whitespace) can be effectively used for enabling IoT devices.

Another most challenging aspect for the network backbone will be the ability to handle this burst of IP addresses and, at the same time, manage and apply the data acquired.

For example, how the notification for a blood pressure level from a patient can be distributed to doctors to provide better life-saving services. How effective navigation can enhance the user-experience for in-vehicle infotainment using Internet access nodes for real-time traffic updates. Or, how smart sensors can be used in a building to switch off lights in unused rooms to save energy.

The amount of notification generated by these sensor-based nodes will be huge. How best the communication network can stream/store in cloud and apply analytics engines to extract key information to benefit end-user needs will determine the extent to which service providers can monetise their investment.